Reimplementing "Interpreting Indirect Answers to Yes-No Questions in Multiple Languages"

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1 Introduction

1.1 Task / Research Question Description

When was the last time you received a simple, straightforward yes or no as an answer to a yes-no question? Humans often tend to give out some context, reasoning, or indirect form of responses to even the simplest questions. Sometimes instead of giving out a simple yes-no, they anticipate the follow-up question and answer to that question, which obviously answers the initial yes-no question but also provides additional and sometimes unwanted information.

This is the basis of the very task that the paper "Interpreting Indirect Answers to Yes-No Questions in Multiple Languages" (Wang et al., 2023) aims to solve. It aims to interpret these indirect answers to simple yes-no questions and provide a simple response that is either yes, no, or maybe. It aims to eliminate the extra clutter that came along.

The paper also aims to solve this task in not a single language only but also provide multilingual functionality.

1.2 Motivation and Limitations of existing work

Multilingual question-answering tasks have been in the hunt since the very beginning but have received importance and much-needed hype recently. There has been a significant amount of research and work being put into solving this specific task of interpreting responses from indirect answers.

The state-of-the-art models such as XLM-E achieve only 68% to 76% F1-score on multilingual Question Answering benchmarks such as MLQA and XQuAD. This means that there is room for improvement in the performance of these models. LLMs like Instruct GPT achieved promising results with the English language. Open-source LLMs such as LLaMa are pre-trained on Latin or Cyrillic scripts and thus have limitations with other scripts.

The majority of these approaches lacked one common feature which is multilingual capability. Almost all of these models were either trained to perform only in the English language or didn't have enough training data in other languages to address multilingual capabilities. Even if they did, the F1 scores were far off from satisfactory.

1.3 Proposed Approach

First, the paper presents a distant supervision approach to collect training data in various languages (Hindi, Chinese, Korean, Spanish, Turkish, Nepali, Bangla, and Persian).

Secondly, it talks about how polar keywords can impact the interpretation of a sentence. Turns, out that a few keywords can give us the intent of the entire sentence.

Finally, it provides 3 approaches to training and fine-tuning the collected data for better performance in various languages. These are called - 1. Cross-Lingual Learning from English, 2. Fine-tuning with the New Language, and 3. Fine-Tuning with Several Languages.

There are four main contributions of this paper, which are as follows :

1. A distant supervision approach to collect yes-no questions with direct answers along with their interpretations as training data and that too in five languages.

2. Setting evaluation benchmarks in eight different languages.

3. Experimental results show that training with the direct answers obtained via distant supervision is beneficial to interpreting indirect answers in the same language.

4. Experimental results show that multilingual training is beneficial, even when no additional training data for the language of interest is available.

1.4 Likely challenges and mitigations

The task of understanding the meaning of answers to the yes-no questions given in two different languages is associated with several difficulties and limitations. Some of the limitations that the authors faced are that, although the F1 score was ensured through double-annotation, only five out of eight languages in the benchmarks underwent this process because of the difficulties in recruiting native speakers for the remaining three languages (Bangla, Nepali, and Persian) thus, there was no reported inter-annotator agreements and can result in the reliability of the data to be affected.
Furthermore, while the choice of three labels (Yes, No, and Middle) for answer interpretations enables clarity, it may deviate from more intricate label sets in some previous studies which could be another limitation in capturing a variety of interpretations.
Also, the experiment is done with the two models XLM-RoBERTa and XLM-Align as the substitutes InfoXLM and XLM-E either give the results present in the benchmark or the models are not open-source. Exploring novel fine-tuning strategies or configurations for XLM-RoBERTa and XLM-Align within the existing data constraints could optimize model performance and enhance the reproducibility of the findings.

2 Related Work

Lal et al. (2022) emphasizes the implicit and underlying reasons that lead to the direct questions, thus, revealing the indirect responses. This investigation not only reinforces our understanding of indirect answers but also enriches our knowledge. On the other hand, while Lal et al. focus on the reasons behind the questions, our study rather concentrates on the interpretation of indirect answers in the particular domain of yes-no questions across different languages, thus making the analysis of the topic more widespread.

Clark et al. (2019) put forward BoolQ; a dataset that contains yes-no questions from search engines and is considered an important resource for the training and assessment of question-answering systems. They are important in the sense that they make the research more complete and more detailed. Instead of focusing on the creation of datasets, our study will analyze and interpret indirect answers to yes-no questions within the realm of different languages. The priority will be given to the understanding and analysis of those answers.

Devlin et al. (2019a)) present mBERT which aids in cross-lingual text understanding and forms the basis for multilingual natural language processing jobs. Their work is not only of enormous importance in the field of cross-lingual research but has also become a solid basis for future studies. However, Devlin et al. concentrated on multilingual language models, while we are looking at the direct translation of indirect responses to yes-no questions across languages. These can be done through distant supervision and cross-lingual learning which do not require human annotations to be made.

3 Experiments

3.1 Datasets

The datasets employed in the initial experiment are Circa, SWDA-IA, and Friends-QIA for English corpora and those acquired through the distant supervision technique for the five other languages (Chinese, Spanish, Hindi, Korean, and Turkish). The distant supervision approach consists of identifying large amounts of unannotated corpora in the target languages and developing the rules that would be used to extract yes-no questions, direct answers, and their interpretations.
These datasets are publicly available. The links to access the used datasets have been provided by the authors of the paper themselves. They are available in a GitHub repository, which is public.

Also, to be noted here is that the authors, of the original experiment, themselves have split the data into Training, Validation, and Testing datasets and the same splits have been used to recreate the models.

Data obtained via distant supervision made available by the authors of the original paper at https://github.com/wang-zijie/yn-question-multilingual

3.2 Implementation

Repo: https://github.com/kkvahd/yn-question-multilingual.git

We forked the authors' GitHub (https://github.com/wang-zijie/yn-question-multilingual), which only consisted of the datasets, and we added our own implementation in the "Implementation.ipynb" file. We also added a "requirements.txt" file and a "helpers.py" file that we fetched from Homework 3 of this course and modified it according to our needs.
It is important to note that we derived some of the code from Homework 3 of this course. Especially the validation performance and training performance code. We, however, modified them according to our requirements.

3.3 Results

Before presenting the results, it's important to clarify our objective: To reimplement and achieve the F1 scores in tables 6, 7, and 8 of the paper "Interpreting Indirect Answers to Yes-No Questions in Multiple Languages", we carried out an experiment to show how well the method performs in different language environments and then to prove that by our re-implementation efforts, we have improved F1 scores. The table below shows the F1 scores development corresponding to the different training approaches and datasets.

Language	Expected	Obtained
Hindi	0.33	0.445
Korean	0.60	0.533
Chinese	0.36	0.381
Bangla	0.45	0.329
Turkish	0.55	0.441
Spanish	0.43	0.525
Nepali	0.53	0.466
Persian	0.55	0.458

Table 1: Results (F1) obtained with XLM-RoBERTa when trained with Circa and SWDA-IA

Language	Expected	Obtained
Hindi	0.43	0.575
Korean	0.62	0.637
Chinese	0.43	0.452
Turkish	0.60	0.568
Spanish	0.46	0.525

Table 2: Results obtained in multiple languages with XLM-RoBERTa, trained blending with English corpora.

Language	Expected	Obtained	Δ F1
Hindi	0.49	0.529	0.084
Korean	0.63	0.595	0.062
Chinese	0.55	0.529	0.148
Bangla	0.55	0.491	0.162
Turkish	0.61	0.541	0.100
Spanish	0.49	0.528	0.003
Nepali	0.62	0.579	0.113
Persian	0.56	0.533	0.075

Table 3: Results obtained with XLM-RoBERTa, trained blending with English corpora and the instances obtained with distant supervision in other languages. Here Δ F1 represents the improvement of F1 score from Table 1.

The tables show the expected and obtained F1 scores across different languages. The scores demonstrate the model's performance when it was trained with the Circa and SWDA-IA datasets, together with English corpora, and with distant supervision instances in other languages. The comparison of the expected and obtained F1 scores shows that the incorporated techniques are responsible for the gradual improvements. The results obtained here not only verify the original findings but also illustrate the progress made in interpreting the indirect answers to yes-no questions in multilingual settings. They not only prove the effectiveness of our re-implementation efforts but also give us helpful clues for further research in this field.

3.4 Discussion

Overall the reproduction was challenging with a few issues, which were addressed to our satisfaction levels. To start with, we did not have access to the exact code the authors used. We also didn't have any reference code or anything from the authors that we could have used to build upon.
On the bright side, we did have access to the datasets, which helped a lot, and would have been impossible to reproduce the results without it. Next, the authors also mentioned the model and the transformers they used, which helped us.
Not having access to the source code, helped us challenge ourselves to develop the code from scratch and make it work. I would say we were successful in this to a great extent.
The results obtained by us were almost similar, not much of a difference from what the authors achieved. We started with a big difference in results, mostly because there was no mention of the

exact hyper parameters the authors used in training their model. We were experimenting and playing around with it for days, just to get the right combination of them which would give us the required results. In the appendix of the paper, the authors did mention a few of them, like maximum no of epochs, minimum learning rate, minimum batch size, etc. which kind of gave us some benchmarks or limits to reduce our experimenting sample size. For almost most of them, we achieved the expected results of the authors. Exceeding on a couple of occasions as well. All the main point results (table 3) reproduced are at a maximum difference of 0.04. Which if further worked upon can be reduced further and reach the expectations. We will be continuing our efforts in this and for checkpoint 2, we will have the exact results reproduced with the right set of hyper parameters.

3.5 Resources

The resources used to reproduce the results mentioned in the paper were pretty significant. It took us around 1 week to select a suitable paper and in our interest area. It took 3 people around 1 week to write the code in the Python notebook. The training for the model using different hyper parameter combinations to get the same result as the authors involved a lot of experimentation.

The authors mentioned the use of a single NVIDIA Tesla V100 (32GB) GPU to do the experimentation. We were not lucky enough to get access to the said GPU. Instead, we made use of T4 GPU on Google Colab to reproduce this. Also, we had to take the pro version of Google Colab to get continuous access to the GPUs. We used around 250 computational units provided by Colab Pro (Paid) and will be requiring some more to implement our plan for checkpoint 2.

Overall, we made complete use of the allocated time to reproduce the results, around 1 week with 3 people spending roughly around 7-10 hours each to select a paper. Next, we spent around 35-40 hours in total to research the models and develop code to implement it. Next, a lot of time was spent experimenting with the hyper parameters and training the model to achieve the same results if not better. On average, each epoch took around 15-20 minutes to run. We do not have an exact estimate of how much time was spent on this process. We spent almost 10-14 days in this spending around 7-9 hours each day per person and our colab code executing the training loops running almost indefinitely. We also spent a couple of days writing this report, showcasing our efforts and results.

3.6 Error Analysis

The authors have performed and included error analysis in their paper. For starters, the model fails to comprehend a rhetorical question in Persian with negation in it. Next, it faces the problem of ambiguity. It does not understand and gets confused by the term weekend. It considers only Sunday as the weekend. On certain occasions, it fails to understand negations. Also, the model fails to read the context, for example, when someone asks if they have pets the response goes that they are afraid of dogs. The model thinks that their fear is due to the presence of dogs, hence they have pets. But in reality, the respond-er does not have pets due to fear. Another one is when the model assumes an answer (yes-no) when there is uncertainty (middle). The last one is where the model fails to process polar distractors like 'no longer yet'.

Upon considering a few examples, We feel that the authors clearly did not analyze how the model would respond to sarcasm. Humans tend to have a good sense of humor and can use sarcasm in their responses. This was clearly not analyzed. Analyzing sarcasm is not a straightforward task, it requires context-based reasoning and reading the environment. Hence not analysing it is justifiable. Another analysis the authors could have done could be how the usage of certain lingo or slang affects the responses.

We have also done a certain level of error analysis. We went through the wrong output answers in Hindi (the language we understand) and tried to understand the similarity in the responses for which the model gave out the wrong outputs. We found that there were some sarcastic answers and there were some answers that were like telling the question ask-er to stop asking questions because the answer was no or the answer was pretty much obvious. These are kinds of examples that the model failed to pick up on. Below are a few examples of the same from the analysis on Hindi dataset translated into English for better understanding.

1. Q :Are you all ready to extend the lockdown?
   A: I wish this question had been asked to those who are far away from the world of

Twitter and Facebook!
Prediction : Middle
Actual : No

2. Q :Should the government abolish ”reservation”? Do you agree?
A: Only an uneducated person could agree to this!
Prediction : Yes
Actual : No

3. Q :Prime Minister Ji, if we give you a book instead of a bouquet, will you give us good days?
A: Oh man,Isn't it a good day that you are asking the prime minister a question ? What else do you want ?
Prediction : Yes
Actual :No

4. Q : Do you also believe that devotees are ignorant like me?
A: Now let's believe that you are mentally disturbed.
Prediction : Yes
Actual :No

5. Q : Government Should Abolish ”Reservation” Do you Agree?
A: Why do you spend 5 rupees every day to tweet?
Prediction : Yes
Actual :No

4 Plan for Checkpoint 2

For Checkpoint 2 we have a few ideas.
We intend to introduce a new combination of the existing languages which will give us a better performance on a particular language. For example, the authors have specified that training the model in English, Korean, and Turkish gives the best F1 score for Korean data. We want to devise another combination of languages like (Spanish, English, and Korean) that gives out a better performance than the existing combination.

Another plan could be to try and use the existing language datasets to train the model and have it predict answers for another language not mentioned in this paper. This will be contingent upon our getting hold or creating of a dataset that is worthy of the task. Also, we could use the new language dataset to alter the existing language combinations to achieve improved performance. We also plan to utilize data augmentation by translating into a new language and then finding combinations of languages to make improvements. This could potentially help us with robustness and making the language multilingual (further) too
Lastly, we could check for robustness and create a model that works great and is robust to a great extent.

5 Workload Clarification

At this checkpoint, we adopted a strategic approach to distribute the workload among the team members equally. Karthik initially took charge of selecting the paper and initiating the process by recreating the code from scratch, ensuring a solid foundation for the project. Meanwhile, Sagar and Karthik collaborate closely on the crucial task of obtaining F1 scores, dedicating extensive hours to running the code continuously for meticulous analysis. Concurrently, Ayush focused on error analysis and studied the related work, to find the model improvement opportunities. Although everyone in the team carried out a specific task, the team as a whole ensured that the workload was equally distributed by investing the same amount of effort in report writing.

—— PART 2 STARTS HERE ——

6 Approach for Training a Multilingual Model

The paper had a decent framework for collecting data from a specific language. The approach/framework is called “distant supervision”. We decided to collect data in **Italian** language.

We went through Frequently Asked Questions and Community Question Answering datasets from MFAQ (De Bruyn et al. (2021)). This is the same dataset from which the authors extracted the Turkish datasets. The dataset was available on huggingface and was made available by the authors of the MFAQ paper. The Link : https://huggingface.co/datasets/clips/mfaq

6.1 Data Collection and Cleaning

MFAQ had an Italian language dataset which we used for our task. The dataset was already split into train and test data. There were a lot of non-yes/no questions and noisy data in the dataset

which needed filtration. We followed the same approach to clean both the train and test datasets.
We started with around 155,000 records on the train and 2,500 records on the test. This data needed cleaning.
The first filtration to be performed was to identify the yes-no questions from the dataset. The rules for which are as follows:

1. Contains a question mark and any of the following question phrases: Did, will, is, are, can, should, etc.

2. Does not contain the wh-question phrases like: che, cosa, che cosa (what), chi (who), dove (where), come (how), perche (why), Quando (when), Quanto/a/i/e (how much/many), Quale/i (what/which).

3. Does not contain any describe, mention, single word questions.

4. Does not contain any links, mentions, unidentified characters, etc.

Now that we have the questions filtered, it is time to filter the answers.

1. Identify direct answers like: yes, no, definitely, certainly, sure, why not, maybe, not at all, etc.

2. From the indirect answers remove records where the answer redirects to another source, has links, mentions, etc.

3. Does not contain answers that include non-rhetorical questions as responses.

4. Too long answers.

This kind of filters all the data and removes all the unwanted records, which do not suit our task. Just to make sure we have only yes-no questions in our dataset, we back-translated the data to English performed the same filtration again, and removed any unwanted data.
The same process was followed for both the train and test datasets. This brought down the number of records to approximately around 10,000 in the training dataset and 500 records in the test dataset. We simply took out about 10% of the points from the training set to create the validation set. The length of the datasets obtained after processing was similar to the other languages and within the limits of what the authors had advised.

6.2 Data Annotation

Now, we had data that needed labeling. As per the authors' methodology, we annotated the data into 3 labels (yes, no, and maybe). We decided to annotate them individually and see if the questions and their answers were relevant (similar to HW3). We calculated the Raw agreement (= 0.7939814814814815) and the Cohen's kappa score (0.6560386473429951) between them. This proved that we were mostly on the same page and also due to the humongous number of rows in the train data, we decided to distribute them and label them. This took a pretty big chunk of our time.
Following the above process gives us the three different datasets that we need to perform our task. The train, validation, and test datasets that we obtained and used are included in the same GitHub repository. Now we proceed to analyse our model on the Italian dataset that we generated

6.3 Model Training

The first task in analyzing the model involves training the model using the correct set of hyperparameters. We noticed that the hyperparameters did not change across various languages in the paper. So we decided to keep very similar numbers and experiment around them. As mentioned earlier, we used three methodologies to train the model (separately). They were 1) Cross-Lingual Learning from English, 2) Fine-Tuning with the New Language, and 3) Fine-Tuning with Several Languages. These were the hyperparameters of each of the training methodologies:

Parameter	**Method-1**	**Method-2**	**Method-3**
Learning Rate	2e-5	2e-5	2e-5
Batch Size	32	16	16
Max Epochs	10	30	30
Alpha	-	0.2(0.5 for korean)	0.4 - 0.7
Optimizer	AdamW	AdamW	AdamW

Table 4: Parameters used with XLM-RoBERTa when training with Circa and SWDA-IA for the Italian dataset for the 3 Methodologies.

Using the above hyperparameters we trained our model in each of the methodologies the authors introduced. For the second and third methodologies, we experimented with some of the hyperparameters like alpha value (Blending factor), the language of choice, etc. in order to optimize and

achieve the model with the best outcome. The results for the same have been included in the sections below. For approaches 2 and 3, each epoch took around 10-12 minutes. And we mostly used 10 epochs. This means that training a model with a particular methodology took around 2 hours. This was of course increased by a lot because we had to experiment around a lot.

6.4 Italian Dataset Statistics and Benchmarks

The statistics for the Italian training dataset and the benchmarks are similar to what the other languages prescribed by the author have. The distribution is as follows :

	Training	Testing
Instances	10,295	432
% yes	69.8	55.7
% no	22.1	31.5
% maybe	8.1	12.8

Table 5: Distribution of Italian datasets

7 Approach for Analysing a Multilingual Model

One of the primary goals of this paper was to set some baseline benchmarks for multilingual performance. It did it in 8 languages and the results were fairly close. Hence, we knew that if our approach produced scores similar to the other languages that the authors worked on, our approach was okay. We however decided to take it a step further and calculate accuracies too. We felt that even if f1 scores (the metric used by the authors) do not improve considerably, accuracies probably could.

Our comparison of the results is with the average results obtained by the other languages in each approach. Also, we have compared the results of the Italian dataset with the best-performing language from what the authors used. We did the comparison for all 3 experimental methods proposed by the authors.

8 Multilingual Model Performance

As mentioned earlier, we consider the benchmarks created by the authors to evaluate our model. We noticed that the metric values were fairly close and we decided to compare average values for all methodologies and the best values for methodologies 2, and 3 (because they actually are multilingual).

8.1 Cross-Lingual Learning from English Results

Looking into the average scores obtained by the authors and obtained by us for Cross-Lingual Learning from English.

Avg. F1 score of other languages	Best F1 score of other languages	F1 score Italian	Accuracy Italian
0.475	0.55	0.399	0.557

Table 6: Comparing author's average results for Cross-Lingual Learning from English to ours'

Table 6 is really not a great indicator and is not even multilingual because the model was only trained on English data only.

8.2 Fine-Tuning with the New Language Results

Now, Looking into the average scores obtained by the authors and obtained by us by Fine-Tuning with the New Language. The authors' scores are the average scores for 5 languages - Hindi, Korean, Chinese, Turkish, and Spanish. Our scores are only based on the Italian language.

Average F1 scores of other languages	Best F1 score of other languages	F1 score Italian
0.508	0.62	0.68

Table 7: Comparing author's average F1-score for Fine-Tuning with the New Language to ours'

The Blending factor (alpha) which decides the blend of the new language with English for this case is 0.5 achieved through experimentation. The details and results of which are included in the Appendix A.

8.3 Fine-Tuning with Several Languages Results

Finally, Fine-Tuning with Several Languages proved to produce the best results (in general and language-dependent) and hence, we did an in-depth analysis of our scores vs the authors' average scores.

Average F1 scores of other languages	Best F1 score of other languages	F1 score Italian
0.56	0.63	0.69

Table 8: Comparing author's average F1-score for Fine-Tuning with Several Languages to ours'

The results obtained are by blending Italian data with Turkish and with an alpha value of 0.67. Experiments were performed to arrive at this combination and the exact set of hyperparameters applied. The details of which are included in the Appendix B.

Method	F1-Score	Accuracy
Cross lingual learning from english	0.40	0.56
Fine tuning with a new language	0.68	0.73
Fine tuning with several languages	0.69	0.70

Table 9: Our best-performing models across the 3 methodologies

In the end, The above shown results are the best F-1 score and Accuracy that we were able to achieve from performing the experiment with all 3 methods. The results for method-3 i.e. Fine tuning with several languages using Italian exceed what we have achieved for any other language.

9 Analysis of Model Multilinguality

9.1 Analysis of Model's Performance

Comparing our model's scores to the benchmarks created by the authors proves that we produced a good dataset and a good model. Now, here are the performances of our model using the 3 methodologies.

For cross-lingual learning from English, which is our first methodology where the training is in English but tested on other languages as well. For Italian data, the results are pretty average and below expectations. This proves one of the points that training only in English language is setting up for failure in other languages. Hence, it does not help our cause.

Coming to our second approach where we Fine tune with a new language (Italian). The model performs pretty good. It stands out as the best language among all others. This methodology improves the performance over the dataset from the previous approach by great means.

There is another layer to method 3 i.e. Fine-Tuning with Several Languages. We used a greedy approach, similar to the authors to create a variety of blends. The best-performing languages (individually for 1 epoch) were Hindi, Turkish, and Spanish. Hence, we decided to combine them and train the model (this is visible in the code that we provided). After experimenting, we concluded that the Turkish dataset had the best performance. Here is the analysis of the same. Even here, when we blend Italian data with Turkish data and train, the performance improves and stands out as the best performing language blend across all other languages. The increase in the performance from the previous 2 methodologies also proves the point that Fine tuning using similar languages improves the performance of the model.

It was surprising to see that the blend of Turkish with Italian produced the best results even though they are not related. It was expected that Spanish would be helpful and support Italian. Although it showed potential but when trained and evaluated it fell flat and didn't come through as the best one. Even the blend of Spanish and Turkish with Italian stood out as a promising and ideal choice owing to them being the best performing languages individually but when combined together did not do as good as Turkish alone.

Testing our Italian + Turkish blend model on other languages We realize that the models are language-specific but we decided to test our model Fine-Tuned in English, Italian, and Turkish language on other languages too. It is in fact a good performance. The results come close to the respective language blended training. Which proves that the model's performance is at par. The results are included in Table 10.

Language	F1 - Score
Hindi	0.44
Korean	0.60
Chinese	0.54
Bangla	0.46
Turkish	0.53
Spanish	0.54
Nepali	0.57
Persian	0.53

Table 10: Results (F1) obtained with XLM-RoBERTa when trained with a blend of English, Italian, and Turkish datasets

9.2 Error Analysis

Obviously, our model made a bunch of mistakes and we decided to analyze them. We noticed that there are answers with a certain condition attached to them. The model fails to identify these conditions while producing the answers.

1. Q: I have already paid off my open positions with the previous supplier and have now paid the Cmor commission to Edison Energia. Will I receive a refund?
   A: This will happen only after your payment is registered by your previous supplier and communicated to Edison Energia
   Prediction: No
   Actual: Yes

2. Q: Can I decontaminate the entire car with one kit?
   A: Enough for a mid-size car!
   Prediction: Yes
   Actual: Middle

The model also fails to identify answers that give a solution instead of addressing the possibility.

1. Q: I forgot my debit card, can I still withdraw?
   A: Enter the app, on the homepage, you will find the "withdraw" option, this option allows you to withdraw for free at UniCredit BancoSmart ATMs.
   Prediction: No
   Actual: Yes

2. Q: Is there a dress code?
   A: Comfortable and casual clothing is recommended.
   Prediction: Yes
   Actual: Middle

Additionally, as mentioned in our previous analysis and the presentation, the model fails when the questions/answers are ambiguous, sarcastic, or rhetorical. Also, a lot of times, the model lacks context and domain-specific knowledge and hence, fails to predict correctly.

9.3 Where our model performs well

As expected, our model does well when we have clear yes/no answers for instance.

1. Q: Hi, is it possible to use this product with refillable spray cans?
   A: The product must be sprayed with the classic spray bottle and trigger, it does not go in a can
   Prediction: No
   Actual: No

2. Q: Is it waterproof?
   A: The glove is not waterproof.
   Prediction: No
   Actual: No

We also noticed that our model does well when there are many complex instructions and the answer ends up being "middle".

1. Q: Is it suitable for classic "soft' BMW/VW/Audi dashboards?
   A: The product is suitable for all skin types, but if the dashboards are made of plastic you can use the 3in1 Dashboard Treatment
   Prediction: Middle
   Actual: Middle

2. Q: After reopening my buddybank account, can I invite friends and get bonuses?
   A: If you have already been a buddybank customer, when you reopen your current account you will not be able to receive the bonus as a guest (see previous question); however, you will be able to invite your friends and receive bonuses as the inviter.
   Prediction: Middle
   Actual: Middle

10 Discussions for checkpoint2

We found out that one of the most critical parts of this experiment is finding the right kind of data. We need very specific and constrained data that would usually not be available easily. After finding the data, having the data be correctly labeled

also seemed to be challenging as manually labeling the amount of data needed to properly train the model is very time-consuming.

The experiment also gave us the insight that once all the data is properly labeled, training the model for the final method with all the different combinations of the existing languages is also a very brute-force approach. This could sometimes lead to training the model on combinations of languages that produce very subpar results and be inefficient, as training the model itself takes a very long time.

We believe that our dataset is good, we could have improved the quality of our Italian train set if we had more time. As mentioned earlier, the train data could not be verified again and we distributed and labeled the data among ourselves. If we had time, we could have had native speakers of the language annotate for us, which would have improved the quality of data. We also could have experimented more with the model and its parameters further if we had more time. Because of the brute force nature, each epoch took around 10-12 minutes and correct analysis took a lot of time.

We see that for fine-tuning with several languages, there is a scope of improving the blends that we create. Maybe there is a better approach than the brute force solution that the authors suggested. Simply, experimenting and playing around with different combinations costs resources, time, effort and money which can be minimised if there was another efficient approach. We failed to identify or design that approach, but hope that in the future it can be done.

11 Workload Clarification for checkpoint2

We adopted a very similar approach to Checkpoint 1. Initially, Sagar focused on finding, cleaning, and modifying the Italian dataset with help from others, while Karthik and Ayush worked on annotating the test data in Italian. We all collaborated on annotating the training data and utilized it to train the model and evaluate its performance on the test data. Each one of us put in the effort and hours to try different combinations of languages and look for the one with the better performance. Additionally, we collectively contributed to crafting the final report for the project.

12 Conclusion

Part1:
Yes, the paper is very much reproducible. The authors have provided a great description of what has been used and what has been done. They have provided the datasets, specified the models to use, and provided a clear picture of what the paper is trying to prove. All this makes the task of reproducing the results favorable. By using the datasets provided along with the models to be used, we have developed our entire code from scratch taking a few references, and were able to replicate the results. Although, we did face difficulty in getting the exact results. Owing to the fact that the exact hyper-parameters used for training have not been mentioned in the paper. Also, the availability of computational resources like GPUs made us take more time and put in more effort in replicating it. Overall we feel that we did the best possible reproduction of the paper.

Part2:
We conclude that the model can be made multilingual. The approaches for data collection and creation combined with the training methodologies make it possible to produce good results with a brand-new language. It can be re-implemented and additional languages can be added on. For this part, we introduced a new language which is Italian. We worked on getting the datasets and processing them. Then we proceeded to train the model and analyse the results. We compared the results on Italian language with other languages and it came out pretty good. For the future, we could add other languages and analyse the performance on them. Work can be done on having a method or process to obtain the language blend, instead of going with brute force and greedy experimentation.

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APPENDIX

A. Fine tune with new language

The value of alpha has been achieved through experimentation with different alpha values and the results of that experiment are included below in the table.

Alpha value	F1 score	Accuracy
0.2	0.6395	0.6968
0.5	0.6849	0.7292
0.7	0.6670	0.6829

Table 11: Comparing the F1-score and Accuracy for Fine-Tuning with the New Language across different alpha values

B. Fine tuning with several languages

Language	F1-Score	Accuracy
Hindi	0.670	0.703
Korean	0.610	0.703
Chinese	0.651	0.708
Turkish	0.703	0.706
Spanish	0.683	0.7152
Turkish + Spanish	0.654	0.701

Table 12: Updated Results (F1-Score and Accuracy) from Fine - Tuning Italian With Several Languages

From initial analysis Turkish and Spanish prove to have have the best scores. We tried blending turkish and spanish for the 3rd experiment and got the F-1 score and accuracy as shown in Table 12. Although they are high, they are not higher than what we had for just training with Turkish

We has also tried blending Italian with other languages in multiple combinations using the greedy approach, but we got the best F-1 Score and accuracy when training with Turkish as shown in Table 13.

Blend	P	R	F1	A
I+S+T	0.7	0.7	0.66	0.7
I+S+T+H	0.66	0.68	0.65	0.68
I+S	0.64	0.64	0.63	0.68
I+T	0.68	0.7	0.69	0.7

Table 13: Analysis of the blends that we used. Italian(I), Spanish(S), Turkish(T), and Hindi(H)

C. Output screenshots

The function get_validation_performance_part2() outputs the parameters in this order: f1 score, accuracy, precision, recall.

```
seed_everything()
get_validation_performance_part2(italian_test_set)

Num of correct predictions = 241
/usr/local/lib/python3.10/dist-packages/sklearn/met
  _warn_prf(average, modifier, msg_start, len(resul
(0.39954460403940345,
 0.5578703703703703,
 0.3112193501371742,
 0.5578703703703703)
```

Figure 1: Output for Cross-Lingual Learning from English on Italian dataset

```
======== Epoch 10 / 10 ========
Training...
Total loss: 90.55054352559853
Num of correct predictions = 68
Validation accuracy: 0.7727272727272727

Training complete!
0.7727272727272727

[ ] seed_everything()
    get_validation_performance_part2(italian_test_set)

Num of correct predictions = 315
(0.6848831632058171,
 0.7291666666666666,
 0.7000570776255707,
 0.7291666666666666)
```

Figure 2: Output for training and testing "Fine-Tuning with the New Language" on Italian dataset

```
======== Epoch 10 / 10 ========
Training...
Total loss: 80.20948643622695
Num of correct predictions = 63
Validation accuracy: 0.7159090909090909

Training complete!
0.7159090909090909

[ ] seed_everything()
    get_validation_performance_part2(italian_test_set)

Num of correct predictions = 303
(0.654741332897673, 0.7013888888888888, 0.6984411272483598, 0.7013888888888888)
```

Figure 3: Training and testing "Fine-Tuning with the multiple languages" on Italian dataset with blend of Spanish and Turkish datasets

```
======== Epoch 10 / 10 ========
Training...
Total loss: 82.8874709340604
Num of correct predictions = 58
Validation accuracy: 0.6590909090909091

Training complete!
0.6590909090909091

[ ] seed_everything()
    get_validation_performance_part2(italian_test_set)

Num of correct predictions = 294
(0.6605096177008567,
 0.6805555555555556,
 0.6797769238063954,
 0.6805555555555556)
```

Figure 4: Training and testing "Fine-Tuning with the multiple languages" on Italian dataset with a blend of Hindi, Spanish and Turkish datasets

```
======== Epoch 10 / 10 ========
Training...
Total loss: 86.3007308832913
Num of correct predictions = 60
Validation accuracy: 0.6818181818181818

Training complete!
0.6818181818181818

[ ] seed_everything()
    get_validation_performance_part2(italian_test_set)

Num of correct predictions = 295
(0.6292668611148218,
 0.6828703703703703,
 0.6427027022913776,
 0.6828703703703703)
```

Figure 5: Training and testing "Fine-Tuning with the multiple languages" on Italian dataset with a blend of Spanish dataset

```
======== Epoch 10 / 10 ========
Training...
Total loss: 88.74146721145148
Num of correct predictions = 61
Validation accuracy: 0.6931818181818182

Training complete!
0.6931818181818182

[ ] # saving the model here to load it later on and per
    torch.save(model, '/content/Italian_Model_Multiling

[ ] seed_everything()
    get_validation_performance_part2(italian_test_set)

Num of correct predictions = 300
(0.6853297686581499,
 0.6944444444444444,
 0.6835561513097745,
 0.6944444444444444)
```

Figure 6: Training and testing "Fine-Tuning with the multiple languages" on Italian dataset with a blend of Turkish dataset